

Carlson Zheng

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SKILLS & CERTIFICATES

Programming: Python, C/C++, Java, MATLAB, Bash, VHDL, Assembly

Frameworks/Libraries: Flask, OpenCV, Tensorflow, Pandas, scikit-learn, numpy, mediapipe

Developer Tools: Git, STM32, Arduino, ESP32, KiCad, Linux, Visual Studio, Analog Discovery 3, LTspice

Certificates: Machine Learning Specialization, Unsupervised Machine Learning, Advanced Learning Algorithms, Supervised Machine Learning — Issued by Coursera, Stanford Online, DeepLearning.AI

EXPERIENCE

Control Systems & Motherboard Team Member

Dec 2025 - Present

Alberta Bionix

Edmonton, AB

- Led a comprehensive hardware study in compute architectures such as MCUs, FPGAs, and DSPs, to select an optimal processing platform based on latency and power requirements.
- Reduced CPU load by 30% by offloading encoder and IMU data acquisition to hardware accelerators, preserving critical processing bandwidth for control loops.
- Designed schematics in KiCad capturing motor controller logic, BLDC circuitry, PSU distribution, and PC communication lines, for a custom motor controller PCB.

Co-Founder/Co-President

Oct 2022 - Jun 2024

Scarlett STEM

Calgary, AB

- Co-founded a student-led STEM organization, scaling membership to 20+ students and securing resources to engage members in hands-on engineering and science related activities.
- Structured creative coding sessions using Python and Java to mentor students in programming concepts such as object-oriented software design and data structures, increasing attendance rate by 20%.

PROJECTS

iCANSpeed | C++, ESP32, Arduino, Git, MCP2515

- Stabilized sensor telemetry for a two-node embedded CAN bus network by deploying a low-pass filter in C++ firmware, eliminating signal fluctuations and reducing electrical noise by 65% and <3ms response time.
- Compensated for ADC input swing limits by implementing a 1.5% endpoint deadband to mitigate analog potentiometer drift, guaranteeing zero throttle idle and accurate OLED speedometer readings.

Domain Detector | TensorFlow, Python, Git, Linux, OpenCV

- Developed a gesture recognition system in Python using OpenCV and mediapipe, that identifies specific hand signs by tracking 21 skeletal landmarks.
- Implemented a lightweight neural network model using a 80/20 training split with Tensorflow and Pandas, resulting in a sign accuracy of over 93%.

PomoRobo | Git, C++, Arduino

- Engineered embedded C++ firmware using a finite-state machine to smoothly coordinate a 3-DOF robotic arm across idle, countdown, and active states.
- Resolved mechanical joint jitter using software-driven motion damping and achieved <50ms response latency by refactoring execution loops to interrupt-driven scheduling.

Ted's Clanker | Python, Flask, Render, Git, Discord

- Programmed a multithreaded Python bot combining a Flask web server with the Discord API to reliably automate 500+ monthly scheduled and location-triggered messages.
- Built custom REST API endpoints to update status and control bot tasks in real time, reducing manual configuration time by 30%.

EDUCATION

University of Alberta BSc in Computer Engineering Co-op

2024 - 2029

Alexander Rutherford Scholarship